

LoLDraft: Mathematical Specification for Real-Time Role Inference & Draft Scoring

LoLDraft Algorithmic Engine

pipeline/role_inference.py • ui/app.js

ABSTRACT — This document provides the formal mathematical and statistical specification implemented in `pipeline/role_inference.py`. The system achieves sub-5ms Bayesian role marginalization across locked enemy flex picks, calculates expected-value lane counter and cross-map threat deltas, aggregates allied draft synergies, applies turn-context blind pick vulnerability penalties, and enforces composition damage guardrails against tank resistance stacking.

1. Master Composite Win Rate Score Equation

For a candidate champion c evaluated in assigned role $r \in \mathcal{R} = \{\text{top, jungle, middle, bottom, support}\}$, given locked allies A and locked enemies E , the composite win rate score $\mathcal{S}(c, r \mid A, E)$ is:

$$\begin{aligned} \mathcal{S}(c, r \mid A, E) = & w_{\text{base}} \cdot \text{WR}_{\text{base}}(c, r) \\ & + w_{\text{lane}} \cdot \mathbb{E}[\Delta_{\text{lane}}(c, r \mid E)] \\ & + w_{\text{threat}} \cdot \mathbb{E}[\Delta_{\text{threat}}(c, r \mid E)] \\ & + w_{\text{synergy}} \cdot \sum_{a \in A} \Delta_{\text{synergy}}(c, a) \\ & + \mathcal{C}_{\text{comp}}(c, r \mid A, E) \\ & - \Omega_{\text{blind}}(c, r \mid E) \end{aligned}$$

Calibrated Empirical Weights

The parameters w were calibrated via Ridge Logistic Regression across 50,000 High-ELO (Emerald+) ranked solo queue matches:

Table 1: Calibrated Empirical Feature Coefficients

Coefficient	Calibrated Value	Description
w_{base}	1.00	Baseline win rate prior for champion c in role r .
w_{lane}	1.35	Direct head-to-head lane opponent matchup delta.
w_{synergy}	0.85	Allied draft pairing synergy delta with locked teammates.
w_{threat}	0.60	Cross-map and teamfight threat from off-lane enemies.
w_{blind}	0.90	Penalty scale applied when picking into unrevealed opponent.

2. Exact Bayesian Marginal Role Inference

Modern League of Legends drafts involve frequent flex picks (e.g., Gragas, Swain, Yasuo). Because champion select reveals champions without lane tags, the engine computes exact marginal probabilities $P(e = r)$ over all legal assignments.

Role Priors with Off-Meta Floor

Let $p_0(e, r)$ be the historical pick rate of champion e in role r . To preserve flexibility for niche picks, an empirical floor $\epsilon = 0.005$ is applied:

$$\tilde{\pi}(e, r) = \max(p_0(e, r), \epsilon), \quad \pi(e, r) = \frac{\tilde{\pi}(e, r)}{\sum_{r' \in \mathcal{R}} \tilde{\pi}(e, r')}$$

Permutation Likelihood over k Locked Enemies

Let $k = |E|$ be the number of locked enemy champions. Let $\mathcal{P}(\mathcal{R}, k)$ denote the set of all ordered k -tuples of distinct roles from $\mathcal{R}$ ($|\mathcal{P}| = \frac{5!}{(5-k)!}$). For each assignment $\sigma = (\sigma_1, \dots, \sigma_k) \in \mathcal{P}(\mathcal{R}, k)$, the likelihood is:

$$\mathcal{L}(\sigma) = \prod_{i=1}^k \pi(e_i, \sigma_i), \quad \mathcal{Z} = \sum_{\sigma \in \mathcal{P}(\mathcal{R}, k)} \mathcal{L}(\sigma)$$

The exact marginal probability that enemy e_i occupies role r is:

$$P(e_i \rightarrow r) = \frac{1}{\mathcal{Z}} \sum_{\substack{\sigma \in \mathcal{P}(\mathcal{R}, k) \\ \sigma_i = r}} \mathcal{L}(\sigma)$$

3. Expected Matchup and Threat Valuation

Direct Lane Counter Delta ($\mathbb{E}[\Delta_{\text{lane}}]$)

The expected lane counter delta integrates head-to-head performance across all enemies weighted by their probability of playing the candidate's lane:

$$\mathbb{E}[\Delta_{\text{lane}}] = \sum_{e \in E} P(e \rightarrow r) \cdot \Delta_{\text{lane}}(c, e, r)$$

where $\Delta_{\text{lane}}(c, e, r) = \text{WR}(c \text{ vs } e \text{ in } r) - \text{WR}_{\text{base}}(c, r)$.

Off-Lane Threat Delta ($\mathbb{E}[\Delta_{\text{threat}}]$)

Enemies in different lanes influence teamfights and map control. Their impact is weighted by their probability of *not* playing in candidate lane r :

$$\mathbb{E}[\Delta_{\text{threat}}] = \sum_{e \in E} (1.0 - P(e \rightarrow r)) \cdot \Delta_{\text{threat}}(c, e)$$

Allied Synergy Sum (Δ_{synergy})

Measures historical win rate variance when champion c is paired with locked ally a :

$$\Delta_{\text{synergy}} = \sum_{a \in A} \Delta_{\text{syn}}(c, a)$$

4. Compositional Guardrails: Damage Skew & Tank Resistances

Core Role Damage Priors

Empirical magic damage priors for unpicked core positions $\mathcal{R}_{\text{core}} = \{\text{mid}, \text{jungle}, \text{top}, \text{bottom}\}$:

$$\pi_{\text{AP}}(\text{mid}) = 0.68, \quad \pi_{\text{AP}}(\text{jungle}) = 0.28, \quad \pi_{\text{AP}}(\text{top}) = 0.24, \quad \pi_{\text{AP}}(\text{bottom}) = 0.05$$

with $\pi_{\text{AD}}(r) = 1.0 - \pi_{\text{AP}}(r)$.

Zero-AP Comp Risk

Let $\mathcal{R}_{\text{open}} \subseteq \mathcal{R}_{\text{core}}$ be unpicked carry roles. If allies lack an AP carry ($\text{pct_magic} < 50\%$) and candidate c is physical ($\text{pct_physical} \geq 65\%$):

$$P(\text{ZeroAP}) = \prod_{r' \in \mathcal{R}_{\text{open}}} (1.0 - \pi_{\text{AP}}(r'))$$

Enemy Tank Armor Stacking Factor

Computed from recognized armor-scaling tanks $\mathcal{T}_{\text{armor}}$ (Malphite, Rammus, K'Sante, Taric, etc.):

$$\Phi_{\text{armor}}(E) = \min \left(1.80, 0.60 + 0.45 \cdot \sum_{e \in E} \mathbb{I}(e \in \mathcal{T}_{\text{armor}}) \right)$$

Composition Adjustment ($\mathcal{C}_{\text{comp}}$)

- **Heavy AD Compounding / Draft Trap:** If $P(\text{ZeroAP}) \geq 0.20$ and $|A| \geq 1$:

$$\mathcal{C}_{\text{comp}} = - \min(4.50, P(\text{ZeroAP}) \cdot \Phi_{\text{armor}}(E) \cdot 3.60)$$

- **Composition Anchor AP Bonus:** If team lacks AP, but candidate c provides AP ($\text{pct_magic} \geq 55\%$) with $|A| \geq 2$:

$$\mathcal{C}_{\text{comp}} = + \min(3.50, \Phi_{\text{armor}}(E) \cdot 2.50)$$

(An identical symmetric formulation penalizes all-AP compositions into Magic Resist stackers $\mathcal{T}_{\text{MR}}$).

5. Turn Context & Blind Pick Penalty

Let $P_{\text{revealed}} = \min(1.0, \sum_{e \in E} P(e \rightarrow r))$. When picking into an unrevealed opponent ($P_{\text{revealed}} < 0.25$):

$$\Omega_{\text{blind}} = \begin{cases} w_{\text{blind}} \cdot V_{\text{blind}}(c, r) \cdot (1.0 - P_{\text{revealed}}), & \text{if } P_{\text{revealed}} < 0.25 \\ 0, & \text{otherwise} \end{cases}$$

where $V_{\text{blind}}(c, r)$ is the historical average loss margin against the champion's top 5 counter-picks.

6. Complete Unified Win Rate Formulation

$$\begin{aligned} \mathcal{S}(c, r) = & 1.00 \cdot \text{WR}_{\text{base}}(c, r) \\ & + 1.35 \sum_{e \in E} P(e \rightarrow r) \cdot \Delta_{\text{lane}}(c, e, r) \\ & + 0.60 \sum_{e \in E} (1.0 - P(e \rightarrow r)) \cdot \Delta_{\text{threat}}(c, e) \\ & + 0.85 \sum_{a \in A} \Delta_{\text{synergy}}(c, a) \\ & + \mathcal{C}_{\text{comp}} - \Omega_{\text{blind}} \end{aligned}$$